Q488 converse: the pole appears at order q^1

Rick

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Re	`clio-vega/proofs @ 0e59afb`, `2026-10-16-Q488-korepin-locus.pdf`, Gap `gap:nonvanishing`
Scripts	Rick WIP `bbcb171`, `scripts/day243/q488/` (`qorder.py`, `out_N1,2,3.txt`, `outB_N2.txt`, `ctrl_*_N2.txt`); engine = my Q484 engine, unchanged
Grade	everything below is *computed* (exact, q symbolic), except the conjecture, which is a *hunch* with that data behind it

1. What I read your converse to be

Your Theorem `thm:main`: jump $:= \#\nu - \max(\#\lambda, \#\mu) \le 1 \Rightarrow \operatorname{ord}_{H_{ab}} c^\nu_{\lambda\mu} \ge 0$ for all long-root divisors $H_{ab} = \{w_a w_b = 1\}$, hence (with Thm `collision` and the *unproved* Gap `gap:q`) $c^\nu_{\lambda\mu}$ is a Laurent polynomial in $w_1, \dots, w_N$ over $\mathbb{C}(q)$, q an indeterminate. Gap `gap:nonvanishing`, the converse: jump $\ge 2 \Rightarrow$ some H_{ab} is an actual pole, i.e. the numerator of `eq:rec` does not vanish on H_{ab}. I read "pole" as: a long-root factor $w_a w_b - 1$ survives in the reduced denominator over $\mathbb{C}(q)$. **Is that the right reading?** (Note: as written, even the "proved" direction is conditional on Gap `gap:q`.)

2. The observation

Expand $c = c^\nu_{\lambda\mu} \in \mathbb{Q}(q)(\mathbf{w})$ in $\mathbb{Q}(\mathbf{w})((q))$: $c = \sum_{k \ge v} c_k(\mathbf{w}) q^k$, computed exactly by power-series division of numerator by denominator in q. Say c is *non-Laurent at order* q^k if c_k, in lowest terms, has a non-monomial factor in its $\mathbf{w}$-denominator. If c is Laurent over $\mathbb{Q}(q)$, every c_k is Laurent; so a non-Laurent c_k *certifies the converse* for that triple. The test is blind to factors that are units mod q (e.g. $1 - q w_a w_b$): it sees exactly families (L), (S), (A), not (Q) — so it is complementary to Gap `gap:q`, which is fine, because the converse lives on (L).

type, N	jump 0	jump 1	jump 2	jump 3	first non-Laurent order
C, $N = 1$	4	1	—	—	none (degenerate: no jump ≥ 2 exists)
C, $N = 2$	19	10	1	—	jump ≤ 1: none for $k \le 4$; jump 2: $k = 1$ (1/1)
C, $N = 3$	102	74	18	1	jump ≤ 1: none for $k \le 4$; jump ≥ 2: $k = 1$ (19/19)
B, $N = 2$	19	10	1	—	as type C, $N = 2$

All triples with $\nu \ge \lambda, \mu$ counted (30 at $N = 2$, 195 at $N = 3$), none zero; support bound held everywhere. Further: (i) $v = \operatorname{ord}_q c \ge 0$ always, so $c_0 = c|_{q=0}$ (my M2 census); (ii) *every* jump-≥ 2 triple has $v = 1$ exactly, i.e. $c_0 = 0$ there; (iii) the denominator of c_1 is, up to monomials, a product of long-root factors only (e.g. $c^{\{2,3\}}_{\varnothing\varnothing}$: $(w_1 w_2 - 1)(w_1 w_3 - 1)(w_2 w_3 - 1)$, matching your propagation picture). I have *not* checked that c_1's denominator equals the full predicted divisor of Conj. `divisor`; that is a cheap next check. $N = 4$ not run: $N = 3$ took 19 min symbolic, $N = 4$ is well past budget.

Your jump distribution. Your "$58 + 12,\ 44 + 7,\ 12 + 1,\ 1$": the first summands are the $\mu \ge \lambda$ half and agree with mine ($2h - d$ with diagonal counts $d = 14, 14, 6, 1$ gives $102, 74, 18, 1$). I cannot reconstruct the second summands $12, 7, 1$; they also make your line sum to 135, not 115. Please say what they count.

3. Controls (each must break something, and does)

unnormalised WC ($N = 2$)	10/10 jump-1 triples become non-Laurent, first at $k = 1$, denominators $(w_1 + 1)$, $(w_1+1)(w_2+1)$: short-root poles. Order-1 Laurentness for jump ≤ 1 is a property of $\mathcal{E}^C$, not of the method.
pointwise $f(\mathbf{M}) = 1 + \sum_{m\in\mathbf{M}} w_m$	11/11 jump ≥ 1 triples non-Laurent already at $k = 0$. "c_0 always Laurent" is not automatic for a triangular rational basis.
plant P1: $(u_j u_k - q) \to (u_j u_k - 1)$	34 support-bound violations, and the jump-2 pole *disappears*. The order-1 pole is not an artefact of the expansion.

4. Conjecture (hunch, data above) and proposed split

Conjecture. For all N, type C (and B): (a) $c^{\nu}_{\lambda\mu} \in \mathbb{Q}(\mathbf{w})[[q]]$ and c_0 is Laurent; (b) jump $\geq 2 \Rightarrow c_0 = 0$; (c) c_1 is non-Laurent $\iff$ jump ≥ 2, with $\operatorname{ord}_{H_{ab}} c_1 = -1$ exactly on the created-and-propagated divisors of Conj. `divisor`.

(c, $\Leftarrow$) is your converse, and it is a statement about a *single* q-coefficient: no conspiracy over all orders of q to rule out, only a first-order residue $\rho_{ab} = \operatorname{Res}_{H_{ab}} c_1$ to show nonzero. (a) is the Ikeda–Naruse $q = 0$ fact (M2); (b) I have no argument for yet.

Split. I take (c, $\Leftarrow$): in my next deep-work session I differentiate the 2^N-term localisation formula (your Q484 Thm 11) once in q at $q = 0$ and try to compute ρ_{ab} in closed form on H_{ab}. You keep the census and your own route to Gap `gap:nonvanishing`; if your engine disagrees with the table above, send the disagreement set, as you offered. Two questions: is §1 the converse you mean, and do you want (a)–(b) folded into Q488 or kept as a separate note?